

Zeyuan Du

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EDUCATION

Tohoku University

Sendai, Japan

M.Eng. in Robotics (IMAC-G)

Oct. 2026 – Sep. 2028

Nankai University

Tianjin, China

B.Eng. in Automation & B.A. in Philosophy

Aug. 2022 – Jun. 2026

- GPA: 82.61/100 | Rank: 23/64 (35.9%)

Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

Visiting Student | Robotics and Autonomous Systems, Systems Hub

Jun. 2026 – Sep. 2026

PUBLICATION

- Z. Du, Q. Wu, T. Yang, and N. Sun, "Brain-inspired Spiking Neural Network-Based Motion Planning and Adaptive Control Methods for Untethered Soft Crawling Robots," under review at *IEEE Transactions on Industrial Electronics (TIE)*, 2026.
- Z. Du, Q. Wu, M. Li, Y. Ma, C. Peng, T. Yang, and N. Sun, "Design of Novel Untethered Hydraulically Driven Soft Crawling Robots," in *Proc. of the 2025 International Conference on Mechatronics, Robotics, and Artificial Intelligence (MRAI 2025)*, DOI:10.1109/MRAI65197.2025.11135847. (**Best Paper Award**)

RESEARCH EXPERIENCE

Robotic Intelligence and Learning Lab, HKUST(GZ)

Guangzhou, China

Research Intern | Advisor: Prof. Qiang Nie

Jun. 2026 – Sep. 2026

Project: Canine-Inspired Emotional Motion Generation for Quadruped Robots

- Developed a **3D Canine Emotional Space** from real canine motion data to systematically model, characterize, and generate diverse emotion-related expressive behaviors.
- Mapped emotional representations to **Unitree Go2** motions and conducted extensive **real-world experiments** to quantitatively validate emotion expression through natural robotic body movements.

Project: Bipedal Robot Hardware Integration and Control

- Conducted **hardware assembly and disassembly** of a bipedal robot, including mechanical integration, wiring, component inspection, and system troubleshooting.
- Performed **motor configuration, joint calibration, and actuator debugging**, and conducted hardware tests to ensure stable and reliable robot operation.

Project: Learning-Based Robot Boxing and Interactive Control

- Processed **human boxing motion datasets** through skeletal preprocessing and interaction sequence construction to extract coordinated fighting patterns for robotic learning.
- Trained robotic agents to learn **boxing, defensive, and opponent-interaction behaviors** from processed motion data, enabling dynamic and coordinated two-robot sparring in physics-based simulation environments.

Project: Spike-Based Efficient Image Reconstruction with Spiking Neural Networks

- Developed an **SNN-based image reconstruction framework** integrating spike-based image encoding, temporal information processing, and sparse neural computation to reconstruct visual signals from encoded representations.
- Evaluated the trade-off between **reconstruction quality, spike sparsity, and computational efficiency**, exploring energy-efficient visual reconstruction through biologically inspired spiking neural computation.

Underactuated Robotics Laboratory, NKU

Tianjin, China

Research Assistant | Advisor: Prof. Qingxiang Wu

Jun. 2024 – Jun. 2026

Project: Brain-inspired SNN-CPG Control Framework With STDP Learning

- Proposed and implemented a bio-inspired SNN-CPG control framework with STDP learning for real-time adaptive locomotion and obstacle avoidance in untethered soft crawling robots.
- Conducted **simulation and real-world experiments** to evaluate adaptive gait generation and obstacle avoidance, demonstrating improved locomotion performance over conventional CPG control.

Project: Untethered Hydraulically Driven Soft Crawling Robot

- Designed and developed an untethered hydraulically driven soft crawling robot integrating hydraulic actuation, established kinematic models of three-dimensional soft actuators based on the constant-curvature assumption.
- Conducted **prototype fabrication and experimental validation**, evaluating actuator deformation, locomotion performance, and untethered operation under different control conditions.

TEACHING EXPERIENCE

New Oriental Education & Technology Group

Tianjin, China

Senior Physics Instructor T3

Dec. 2025 – May. 2026

- Delivered physics instruction covering mechanics, electromagnetism, thermodynamics, and modern physics.
- Developed customized teaching materials and effective problem-solving strategies for examination preparation.
- Improved students' conceptual understanding and analytical skills through individualized guidance.
- Received strong positive evaluations from students and parents for clear explanations and effective teaching.

Senior Thesis Project Group, Nankai University

Tianjin, China

Teaching Assistant

Sep. 2025 – Jun. 2026

- Coordinated weekly thesis group meetings and organized research progress presentations.
- Assisted faculty in managing team members, scheduling tasks, and tracking project milestones.
- Facilitated communication within the team and helped ensure the timely completion of thesis projects.

AWARDS & SCHOLARSHIPS

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| ● Monbukagakusho Honors Scholarship | 2026 |
| ● National Golden Prize (Top Award) , China Industrial Intelligence Challenge | 2025 |
| ● Completion Certificate , Nankai University Undergraduate "Top 100 Innovation Projects." | 2025 |
| ● 3rd , Innovation Competition of the 7th "President's Cup" of Nankai University | 2025 |
| ● National 3rd , China Engineering Robot Contest and International Open | 2024 |
| ● National 2nd , International Advanced Robot & Simulation Technology Competition | 2024 |
| ● Tianjin Regional 1st , International Advanced Robot & Simulation Technology Competition | 2024 |

SKILLS

- **Language:** Mandarin (Native), English (TOFEL: B2), Japanese (Basic)
- **Software:** MATLAB, C++, Python, PyTorch, Altium Designer, SolidWorks, Abaqus, Arduino IDE, ROS
- **Certificates:** Ascend Developer Certification, Chinese National Chess Player